SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

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Subject: Programming with Java	**Sem: III AY: 2026-27**

Assignment No: 15

TITLE:TITLE :Write a Java program to demonstrate exception handling using finally, throw, and throws.

Problem Statement
Write a Java program to demonstrate exception handling using finally, throw, and throws.

Learning Objectives
To implement finally, throw, and throws for effective exception management.

Theory
The finally block always executes regardless of whether an exception occurs, making it suitable for resource cleanup. The throw keyword explicitly generates an exception, while throws declares exceptions that a method may pass to its caller. These mechanisms improve structured error handling and application stability.

Exercises:

1. Create a Login program that throws an exception for invalid password and uses finally block.

Users > sumit > Documents > java > java1 > Login.java > ...

```java
import java.io.*;

public class Login
{
    static void checkPassword(String password) throws Exception
    {
        if(!password.equals("1234"))
        {
            throw new Exception("Invalid password");
        }

        System.out.println("Login successful.");
    }

    Run | Debug
    public static void main(String[] args)
    {
        try
        {
            checkPassword("123");
        }
        catch(Exception e)
        {
            System.out.println("Error: " + e.getMessage());
        }
        finally
        {
            System.out.println("Login process completed.");
        }
    }
}
```

Output:

```
[Running] cd "/Users/sumit/Documents/java/java1/" && javac Login.java && java Login
Error: Invalid password
Login process completed.

[Done] exited with code=0 in 0.298 seconds
```

2.Create an ATM PIN Verification program that throws an exception for an invalid PIN entered by the user and uses a finally block to display a message indicating that the verification process has completed.

Users > sumit > Documents > java > java1 > ATMPinVerification.java > ATMPinVerification > main(String[])

```java
import java.io.*;

public class ATMPinVerification
{
    static void checkPin(int pin) throws Exception
    {
        if(pin != 1234)
        {
            throw new Exception("Invalid ATM PIN");
        }

        System.out.println("PIN verified successfully.");
    }

    Run | Debug
    public static void main(String[] args)
    {
        try
        {
            checkPin(1111);
        }
        catch(Exception e)
        {
            System.out.println("Error: " + e.getMessage());
        }
        finally
        {
            System.out.println("PIN verification process completed.");
        }
    }
}
```

output:

```
[Running] cd "/Users/sumit/Documents/java/java1/" && javac ATMPinVerification.java && java ATMPinVerification
Error: Invalid ATM PIN
PIN verification process completed.

[Done] exited with code=0 in 0.302 seconds
```

Conclusion:

The programs successfully demonstrate exception handling using finally, throw, and throws in Java. The Login program handles an invalid password, while the ATM PIN Verification program handles an invalid PIN. The throw keyword is used to explicitly generate an exception, and throws declares the exception in the method. The finally block executes after exception handling and displays the completion message. Thus, these programs demonstrate structured and reliable exception handling in Java.